

# TCSR-555

Two-Component Structural Repair

ASTRAL

# TRUBUILD™

TECHNICAL DATA SHEET VER NO. N.1

## PRODUCT DESCRIPTIONS

**TCSR-555** is a two-component, solvent-free, epoxy system for repairing cracked and damaged concrete structures. It is used to bond new structural concrete to old concrete, bricks & steel component as it gives excellent bonding properties with various construction surfaces.



### FEATURES & ADVANTAGES



**EXCELLENT ADHESION**

- It has excellent adhesion and strength on various construction substrates and is easy to apply
- It has very low shrinkage and high open time



**EASY TO APPLY**



**LOW SHRINKAGE**



**HIGH MECHANICAL STRENGTH**

- Easy to apply
- High density
- Free flowing
- High mechanical strength

## AREA OF APPLICATIONS



## AVAILABLE PACKING



**Excellent system to bond old to new concrete/mortar, repair old concrete structures and it can be extended further.**

## METHOD OF APPLICATION

**Surface preparation:** Surface must be thoroughly cleaned. Remove all loose particles, dirt, dust, laitance, oil, grease, etc. by using wire brush, scraping or grit blasting, pressurized air or water jet.

**Holdfast grouting:** Take Part-A, Part-B & silica powder (100 mesh) in 2:1:8 ratio. In a pail, mix Part-A & Part-B in the suggested ratio and mix with a slow speed mixture to avoid entrapping of air. Now charge Part-C and mix for 2-3 min until even colour is obtained. Do the masking of area in which grout is to be applied. Now pour material from one side to eliminate air entrapment.

**Waterproofing:** Take resin and hardener as per mentioned ratio, mix thoroughly, apply the mix on clean and concrete surface. 1 kg material will cover 1 sqmt. area at 1 mm thickness. Leave applied surface untouched for 24 hours. Apply second coat and sprinkle silica coarse sand. After drying, lay 1" thick layer of cement mortar to prevent the treated surface from the effect of ultra violet rays.

**Old concrete to new concrete bonding:** Take resin and hardener as per mentioned ratio, mix thoroughly and apply the mix on old concrete surface. Pour new concrete mix immediately over the area on which TRUBUILD TCSR 555 has been applied. Quartz and silica to be added as per requirement.

**TECHNICAL PROPERTIES**

| S.NO.         | PROPERTIES      | STANDARD       | RESULTS            |
|---------------|-----------------|----------------|--------------------|
| <b>PART-A</b> |                 |                |                    |
| 01.           | Chemical base   |                | Epoxy resin        |
| 02.           | Physical state  |                | Viscous liquid     |
| 03.           | Appearance      |                | Transparent        |
| 04.           | Color           |                | Colorless          |
| 05.           | EEW             | ASTM D-1652-11 | 170-190            |
| 06.           | Viscosity @25°C | ASTM D-4402    | 500-1000 cps       |
| 07.           | Sp. gravity     |                | 1.15               |
| <b>PART-B</b> |                 |                |                    |
| 08.           | Chemical base   |                | Polyamide hardener |
| 09.           | Physical state  |                | Viscous liquid     |
| 10.           | Appearance      |                | Transparent        |
| 11.           | Color           |                | Yellow             |
| 12.           | Amine value     | IS 9162-1979   | 350-400            |
| 13.           | Viscosity @25°C | ASTM D-4402    | 8000-12000 cps     |
| 14.           | Sp. gravity     |                | 0.98               |

| S.NO. | PROPERTIES                                                     | STANDARD    | RESULTS                   |
|-------|----------------------------------------------------------------|-------------|---------------------------|
| 01.   | Mixing ratio                                                   |             | 100:50                    |
| 02.   | Mix viscosity @35°C                                            | ASTM D 4402 | 500-1500 cps              |
| 03.   | Mix viscosity @40°C                                            | ASTM D 4402 | 2000-3000 cps             |
| 04.   | Mix density                                                    |             | 1.12 gm/cc                |
| 05.   | Pot life @ 30°C                                                | Visual      | 150-180 min in 100 gm mix |
| 06.   | Pot life @ 35°C                                                | Visual      | 100-130 min in 100 gm mix |
| 07.   | Hardening time                                                 |             | 12-15 hrs                 |
| 08.   | Curing time                                                    |             | 24 hrs                    |
| 09.   | Application temperature                                        |             | 10-35°C                   |
| 10.   | Shore-D-hardness                                               | ASTM D 2240 | 60-65                     |
| 11.   | Compressive strength (after 24hrs)                             | ASTM C 579  | 45 N/mm <sup>2</sup>      |
| 12.   | Compressive strength with silica in 2:1:8 ratio (after 7 days) | ASTM C 579  | 55 N/mm <sup>2</sup>      |
| 13.   | Lap shear strength (after 24hr)                                | —           | 70-100 kg/cm <sup>2</sup> |

**STORAGE & HANDLING PRECAUTIONS**

Product should be stored in a dry place in their original, properly closed containers at temperature between 2-40°C. Partially emptied container should be closed immediately.

**COVERAGE**

15-20 sqft./kg per 2 coats @ 300 micron

**SHELF LIFE**

24 months from the date of manufacturing

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